

embedding

“river”

$$\begin{bmatrix} 0.2 \\ 0.7 \\ -0.4 \\ 0.9 \\ 0.1 \\ -0.3 \end{bmatrix}$$

+

position 0

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

=

input

“river”

$$\begin{bmatrix} 0.2 \\ 1.7 \\ -0.4 \\ 1.9 \\ 0.1 \\ 0.7 \end{bmatrix}$$

embedding

“bank”

$$\begin{bmatrix} 0.5 \\ -0.1 \\ 0.8 \\ 0.3 \\ -0.6 \\ 0.4 \end{bmatrix}$$

+

position 1

$$\begin{bmatrix} 0.84 \\ 0.54 \\ 0.05 \\ 1.00 \\ 0.00 \\ 1.00 \end{bmatrix}$$

=

input

“bank”

$$\begin{bmatrix} 1.34 \\ 0.44 \\ 0.85 \\ 1.30 \\ -0.60 \\ 1.40 \end{bmatrix}$$

sin and cos
of position 0

sin and cos
of position 1

self-attention

$d_{\text{model}} = 6$ numbers per word,
the same size as the embedding